INCH-POUND

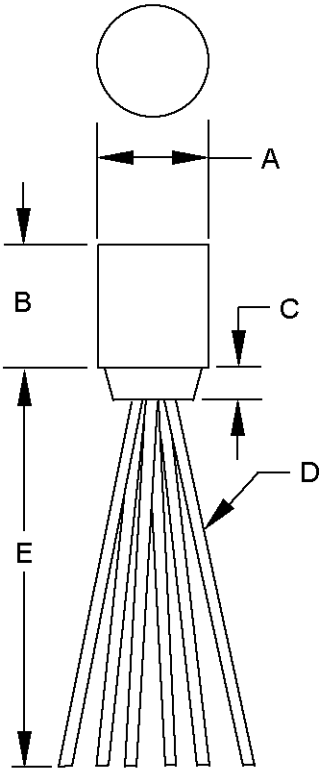
MIL-PRF-27/290C
w/AMENDMENT 2
20 June 2024
SUPERSEDING
MIL-PRF-27/290C
w/AMENDMENT 1
17 March 2017

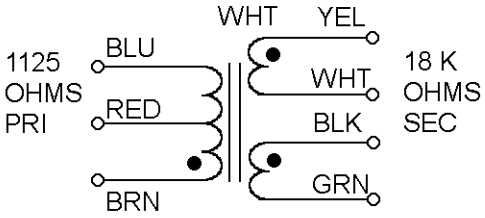
PERFORMANCE SPECIFICATION SHEET

TRANSFORMER, AUDIO FREQUENCY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.





WORKING VOLTAGE: 50 V
ALTITUDE: 150,000 FT

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configurations.

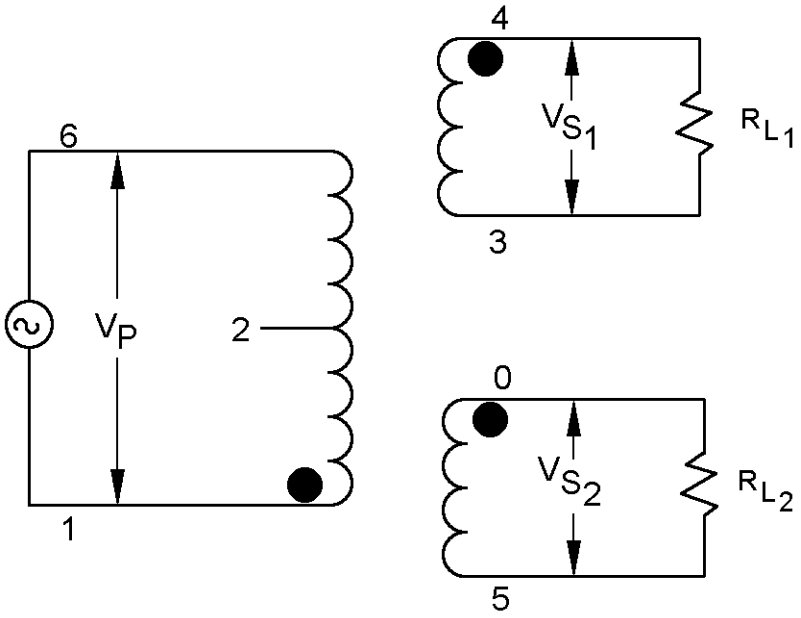
AMSC N/A
FSC 5950

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Dim	Inches	mm
A	.344 Dia. Max	8.74 Dia. Max
B	.469 Max	11.91 Max
C	.030 Max	0.76 Max
D	AWG NO. 26 (.016)	AWG NO. 26 (0.41)
E	1.500 ±.250	38.1 ±6.4

FIGURE 1. Dimensions and configurations – continued.



$V_{S1} = V_{S2}$ ±.05% when:
$V_P = 15 \sin \{2\pi(400 \pm 20)T\}$ where T = Time in sec.
$R_{L1} = R_{L2}$ = 15 kΩ thru 1 MΩ.

FIGURE 2. Output balance circuit.

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REQUIREMENTS: (When colors in parentheses, i.e. (Brn-Blu), are used, they indicate the winding and the extreme terminals of the winding. When the extreme terminals of two windings are used, i.e. (Brn-Blu), the windings are connected in series.)

Electrical ratings:

Primary power level: 100 mW maximum at 1 kHz.

Working voltage: 50 Volts peak.

Primary impedance: (Blu-Brn) 1,125 ohms ct ±125 ohms.

Secondary impedance: (Yel-Grn) 18,000 ohms split ±2,000 ohms. Split indicates that one-fourth of the listed impedance is available by paralleling the windings.

Primary current: 3.5 mA.

Frequency range: 300 Hz to 20 kHz, ±3 dB at 1 mW level.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal, one open end filled with epoxy.

Terminals: Copper clad, steel tinned 100 percent, electroplated per ASTM B545 and ASTM B339.

Diameter: 0.016 inch.

Length: 1.500 ±0.250 inch.

Weight: 0.2 ounce maximum.

Operating temperature range: -55°C to +130°C.

Altitude: 150,000 feet.

Terminal strength: MIL-STD-202-211, test condition A, 2 pounds.

Dielectric withstanding voltage (each winding):

At sea level and at reduce barometric pressure: 100 Vrms.

Polarity: Additive with terminals Blu and Yel, and Wht and Blk connected.

Electrical characteristics:

Primary Impedance: 1,125 ohms ct ±12 percent with 1 volt at 5 kHz and 3.5 mA dc applied to the primary and 18 kΩ across the secondaries.

Primary dc resistance: 120 ohms ±25 percent.

Secondary dc resistance: 940 ohms ±25 percent.

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No load (center tap unbalanced): ±1 percent at 1 volt, 5 kHz across the primary.

Harmonic distortion: Total harmonic content of the output shall be a maximum of 5 percent at 100 mW and 3.5 mA in the primary.

Secondary output balance: When the transformers are operated with identical loads of 15 kΩ to 1 MΩ on each of the secondary windings and the primary input voltage is equal to 15 sin {2π(400 ±20) T}, where T = time in seconds, the output voltages shall be equal within ±.05% (see figure 2).

Insertion loss: At 100 mW in the primary, the insertion loss shall be a maximum of 3.0 dB at 1 kHz.

Frequency response: Z_S = 1,000 ohms; Z_L = 16 kΩ, E_L = 1 volt; reference frequency = 1 kHz.

Frequency range is 300 Hz to 20 kHz, ±3.0 dB.

Self-resonant frequency: Shall be a minimum of 300 kHz.

Vibration (high frequency): MIL-STD-202-204, test condition B.

Marking location: See figure 1.

Part or Identification Number (PIN): M27/290-01.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204 MIL-STD-202-211 ASTM B339 ASTM B545

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2024-022)

Review activities:
Army - AR, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19

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